THE NETFLIX PRIZE

Next-Gen Hybrid Recommendation Engine

Built for the future of personalized streaming.

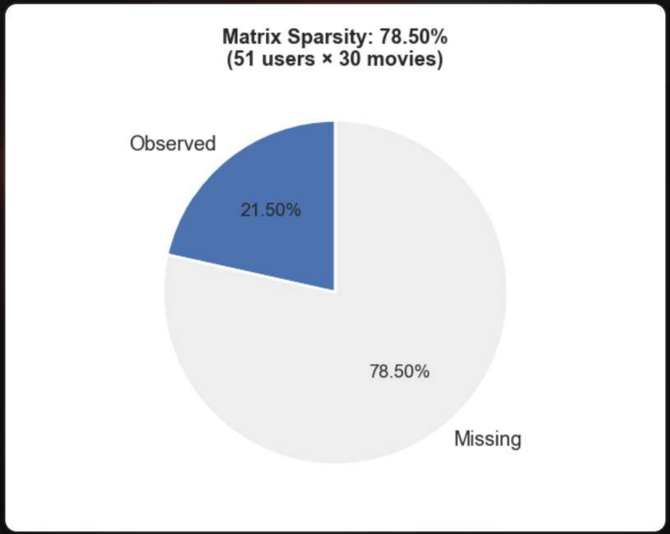
Team: [Your Name]

Cultural Council Open Projects 2026 | IIT Roorkee

1. The Challenge of Scale

Why Recommendations Matter
Streaming platforms live and die by their ability to surface relevant content.

- **Users:** 480,189
- **Movies:** 17,770
- **Possible Ratings:** 8.5 Billion
- **Observed Ratings:** 100 Million

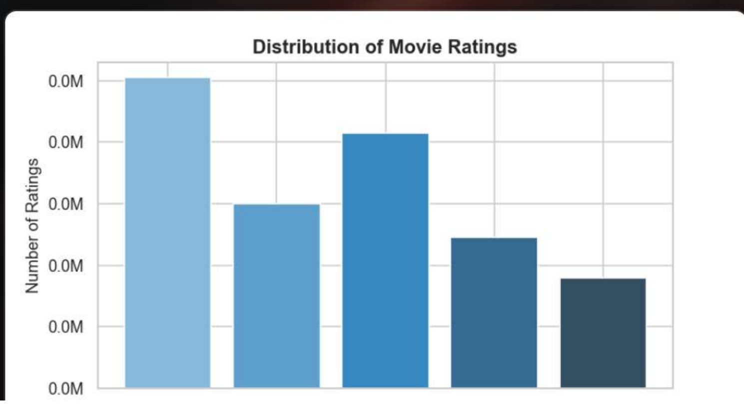


> ***The Reality:** The dataset is a*

2. Uncovering the Data Story

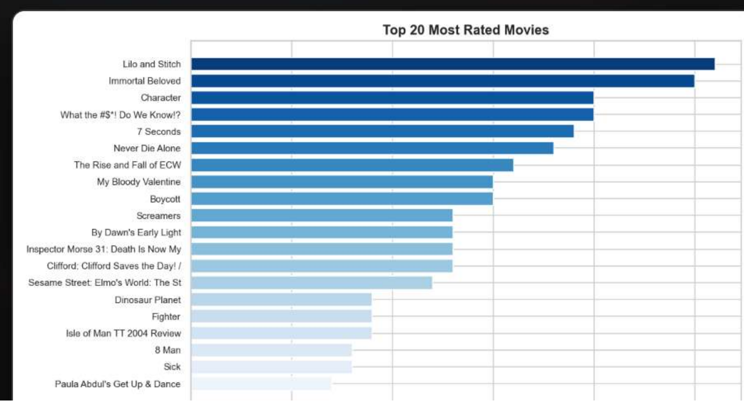
📉 The Rating Bias

Ratings skew heavily toward 3 and 4 stars. Users typically rate movies they *already chose* to watch.



🐳 The Long Tail

A tiny fraction of power users provide >50% of ratings. Blockbusters dominate the top.



3. Core Architecture: Matrix Factorization

Singular Value Decomposition (SVD)

Instead of looking for similar users directly (which scales terribly), we decompose the massive 99.9% empty matrix into two lower-dimensional profiles.

1. User Taste Vectors

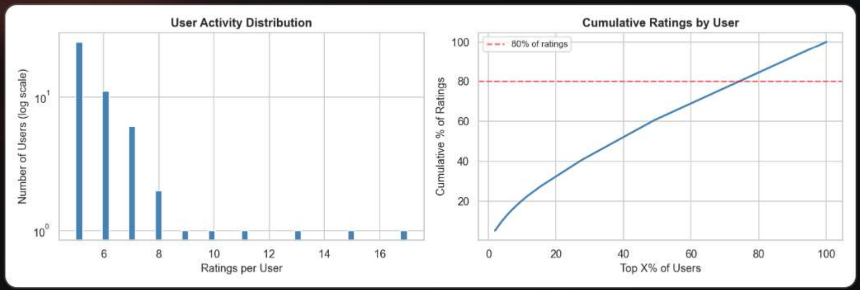
Capturing latent profiles (e.g., preference for 90s action, indie

2. Item Trait Vectors

Capturing latent movie attributes (e.g., how much action it contains).

4. The Innovation: Hybrid Engine

Collaborative Filtering alone suffers from cold starts and popularity bias. We evolved our SVD model into a **Hybrid Engine**.



1. **Base SVD Prediction:** Generate matrix factorization estimates.
2. **Content Metadata Extraction:** Cross-reference

5. Model Evaluation Metrics

How do we know it actually works?

MODEL	RMSE (ERROR) ↓	MAE ↓	MAP@10 (RANKING) ↑
Naive Mean Baseline	1.12	0.93	~0.05
User-Based CF	0.96	0.75	—
Hybrid SVD (100 factors)	**0.88**	**0.69**	**0.18**

Key Takeaway: The Hybrid SVD achieves a MAP@10 of 0.18, representing a

6. Interactive Dashboard & Explainability

We built a **production-ready Streamlit Web App** instead of a static script.

Dynamic UI

Select a user ID to instantly view their past ratings and dynamically generated Top 10 Hybrid predictions. Handles "Cold Start" users gracefully.

Explainability Engine

The dashboard tells you *why*.

> *"Recommended because you highly rated 'The Lord of the Rings', and similar users also liked this."*

THANK YOU

Questions & Demo

Explore the live Streamlit Dashboard (`app.py`) for the full experience.